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Humanoid Parkour Locomotion for Unitree G1 in Isaac Lab

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Abstract

This project builds a custom parkour-oriented velocity-tracking task for the Unitree G1 robot in Isaac Lab. Starting from the official manager-based G1 rough locomotion task, I define three increasingly difficult terrain tiers, train PPO policies with RSL-RL, and evaluate them with survival, traversal-progress, and geometry-aware obstacle-crossing metrics. The results show a clear curriculum effect: easy terrain is stable but narrow, medium terrain is the first tier that transfers strongly to hard terrain, and hard terrain gives the most robust source policy. Ablations further show that reward shaping improves rough-to-hard transfer, while the height-scanner ablation has a non-monotonic effect: no-scan policies remain competitive on moderate stress terrains but collapse to 0% timeout and obstacle crossing on hard stress terrain. Overall, the project provides a reproducible task package and evaluation pipeline for studying humanoid obstacle locomotion.

1 Introduction

Humanoid parkour locomotion requires more than stable flat-ground walking. A useful policy must move over uneven terrain, climb or descend structured height changes, cross gaps, and avoid falls while still obeying a locomotion command. The goal of this project is to build a reproducible parkour-oriented locomotion environment for the Unitree G1 robot in Isaac Lab [1]. The project extends an existing locomotion task with custom terrain design, reinforcement-learning training, systematic evaluation, and ablation analysis.

The repository is a compact Isaac Lab task package. It stores task registration code, terrain configurations, reward extensions, scripts and results. Isaac Lab itself remains an external dependency. Large artifacts such as checkpoints and TensorBoard event files are kept outside the repository.

The project repository is available at: `https://github.com/shenshan222/g1-parkour-isaaclab`.

The main contributions are:

- three custom parkour terrain tiers for Unitree G1, covering stairs, boxes, rough patches, slopes, and gaps;
- a training and play pipeline for easy, medium, and hard parkour tasks using PPO through RSL-RL [3, 2];
- a unified evaluation pipeline with timeout, traversal-progress, and obstacle-crossing metrics;
- a terrain difficulty ablation across flat, rough, easy, medium, and hard settings;

Table 1: **Inherited rough G1 reward terms.** The parkour tasks keep these rough-locomotion terms and add only the parkour-specific terms described below.

Group	Reward term	Weight
Failure	termination penalty	-200.0
Velocity tracking	linear velocity tracking in yaw frame	1.0
Velocity tracking	yaw-rate tracking in world frame	2.0
Gait/contact	positive biped feet air time	0.25
Gait/contact	feet sliding penalty	-0.1
Stability	flat torso orientation penalty	-1.0
Joint posture	ankle joint-limit penalty	-1.0
Joint posture	hip, arm, finger, and torso deviation penalties	$-0.1/-0.05$
Regularization	action-rate penalty	-0.005
Regularization	hip/knee acceleration penalty	-1.25×10^{-7}
Regularization	hip/knee/ankle torque penalty	-1.5×10^{-7}

- a reward-shaping MDP ablation using parkour-progress and foothold-safety rewards;
- a height-scanner observation ablation that removes terrain height-scan input from rough and hard policies;
- an additional ExtremeRandom out-of-distribution stress evaluation for hard and hard-MDP policies.

2 Method

2.1 Framework and task objective

This task is built on Isaac Lab's manager-based Unitree G1 rough velocity-tracking environment [1]. The inherited environment provides the robot model, low-level joint-position actuation interface, proprioceptive observations, height-scan sensing, terrain curriculum machinery, and base-contact failure termination. The project keeps this locomotion interface and velocity-tracking task formulation fixed, and studies how terrain curriculum, reward design, and terrain sensing affect obstacle robustness.

The concrete task objective is to train a policy that drives the robot forward across a generated terrain tile while remaining balanced and tracking the commanded base velocity. In the original Isaac Lab G1 rough training configuration, the command is not a full planar translational velocity: the forward velocity is sampled from `lin_vel_x` $\in [0.0, 1.0]$, the lateral velocity is fixed to `lin_vel_y` $= 0$, and the yaw-rate command is sampled from `ang_vel_z` $\in [-1.0, 1.0]$. The linear velocity command is interpreted in the gravity-aligned yaw frame of the robot base. The parkour environments inherit this command structure.

A successful parkour policy should satisfy three requirements simultaneously: it should avoid falling, track the forward and yaw components of the velocity command reasonably well, and make forward progress across terrain discontinuities such as stairs, boxes, slopes, and gaps.

2.2 MDP formulation

The task is modeled as a finite-horizon Markov decision process

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, r, \gamma, T). \tag{1}$$

The policy observation for s_t is assembled from Isaac Lab manager terms, including base and joint states, projected gravity, commanded velocity, previous action, and terrain height-scan observations. Contact sensors are not part of the policy observation; they are used by reward and termination terms such as foot sliding and base-contact failure. The action a_t is a vector of joint-position targets for the G1 robot. The transition $p(s_{t+1} \mid s_t, a_t)$ is induced by Isaac Sim physics, terrain geometry, contact dynamics, and the command sampler.

The base reward r_t^{G1} is inherited directly from Isaac Lab's rough G1 velocity-tracking task. It is a weighted sum of tracking rewards, gait/contact rewards, stability costs, and action or joint regularization terms. Table 1 summarizes the inherited terms that remain active in the parkour tasks.

On top of these inherited rough-locomotion terms, parkour training adds a forward-progress reward:

$$r_t = r_t^{\mathrm{G1}} + w_p r_t^{\mathrm{progress}}, \qquad r_t^{\mathrm{progress}} = \mathrm{clip}(0.5 v_{x,t}^{w}, 0, 0.75), \qquad w_p = 0.5, \tag{2}$$

Table 2: **Parkour terrain curriculum.** Difficulty increases through higher steps and boxes, wider gaps, stronger slopes, and narrower feasible platforms.

Terrain parameter	Easy	Medium	Hard
Step height	0.05–0.17 m	0.05–0.23 m	0.10–0.28 m
Gap width	none	0.05–0.28 m	0.15–0.45 m
Box/grid height	0.05–0.15 m	0.05–0.20 m	0.08–0.25 m
Slope range	up to 0.30	up to 0.40	up to 0.45

Here, the superscript w emphasizes that the progress term is defined in the world frame. This is different from the inherited linear velocity-tracking reward, which compares the command with the base linear velocity expressed in the gravity-aligned yaw frame of the robot base. In the G1 rough configuration, the lateral command is fixed to zero, so the inherited task mainly tracks forward velocity and yaw rate. The world-frame progress term is therefore not a replacement for command tracking; it is an additional traversal bias that rewards motion along the terrain-forward x direction. When the robot yaw deviates from the terrain-forward direction, this term can trade off against strict command tracking, so tracking error is reported separately in the experiments.

The MDP ablation adds a foothold-safety cost while keeping the robot, observations, action space, terrain generator, PPO algorithm, and evaluation protocol fixed:

$$r_t^{\mathrm{MDP}} = r_t^{\mathrm{G1}} + w_p r_t^{\mathrm{progress}} - w_f c_t^{\mathrm{foothold}}, \qquad w_f = 0.2. \tag{3}$$

The cost is

$$c_t^{\mathrm{foothold}} = c_t^{\mathrm{slide}} + 2.0 c_t^{\mathrm{drop}} + 0.5 c_t^{\mathrm{mismatch}}. \tag{4}$$

Here c_t^{slide} penalizes foot sliding while a foot is in contact, c_t^{drop} penalizes support points that drop far below the base as a proxy for edge or gap stepping, and c_t^{mismatch} penalizes large height differences between simultaneously contacting feet.

The height-scanner ablation changes the observation space. It removes both the height-scanner sensor and the corresponding policy observation term, while keeping the robot, action space, terrain family, PPO runner, and evaluation protocol otherwise unchanged. I train two no-height-scan policies: one on official rough terrain and one on hard parkour terrain.

2.3 Parkour terrain curriculum

The parkour curriculum replaces the official rough terrain generator with three terrain presets. Each preset mixes structured terrain components such as stairs, boxes, rough patches, slopes, and, for medium and hard tiers, gaps. The key difficulty ranges are summarized in Table 2.

2.4 Evaluation metrics

The evaluation pipeline reports three success criteria. Timeout success means an episode reaches the horizon without base-contact failure. Traversal-progress success means the robot avoids failure and reaches at least 4 m forward displacement. Obstacle-crossing success means the robot avoids failure and its base crosses the generated geometry boundary of a supported gap or stairs obstacle. Timeout measures survival, progress distinguishes survival from standing still, and obstacle crossing measures terrain-specific completion.

3 Experimental protocol

The report uses nine trained policies in total: two official Isaac Lab baselines, three custom parkour terrain policies, two reward-shaping MDP variants, and two no-height-scan variants. The main 4-by-4 cross-terrain matrices compare the rough, easy, medium, and hard checkpoints. The flat policy is used as a reference baseline, while the MDP and height-scanner variants are analyzed in Section 5.

The main comparison evaluates how policies trained on different source environments transfer to different evaluation terrains. Each 4-by-4 source checkpoint is evaluated on rough, easy, medium, and hard play environments. I use two cross-terrain protocols: a random protocol that samples terrain type and difficulty inside the terrain grid, and a fixed-row stress protocol that pins all environments to

Table 3: **Baseline comparison.** Rough terrain increases failure rate and tracking error while preserving stable training.

Metric	Flat	Rough
Mean reward	28.77	24.01
Mean episode length	1000.0	984.5
Timeout rate	99.56%	94.62%
Base-contact failure	0.44%	5.41%
XY tracking error	0.2087 m/s	0.3407 m/s

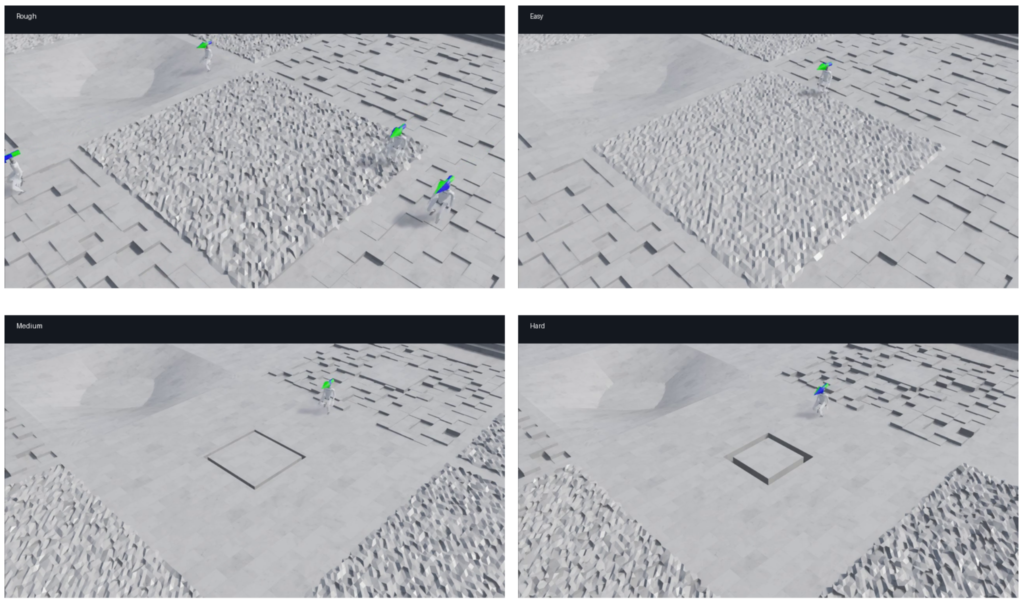


Figure 1: **Policy rollouts on representative terrains.** Frames are extracted from the recorded rough, easy, medium, and hard play videos. They provide qualitative evidence that the learned policies are evaluated on visibly different terrain structures rather than on a single flat locomotion setting.

the highest difficulty row. Unless otherwise noted, each cross/stress evaluation uses 64 episodes, 32 parallel environments, and seed 42.

Detailed task IDs, hardware information, and implementation file locations are listed in Section B.

4 Results

4.1 Reference baselines

Table 3 compares the two official G1 baselines used to validate the project pipeline. The flat baseline verifies the training pipeline: it reaches 99.56% timeout rate and only 0.44% base-contact failure. The rough baseline is harder but remains stable, with 94.62% timeout rate and 5.41% base-contact failure. Its XY velocity tracking error is higher than the flat baseline, increasing from 0.2087 m/s to 0.3407 m/s. This confirms that the official rough task is a meaningful stepping stone between flat walking and parkour.

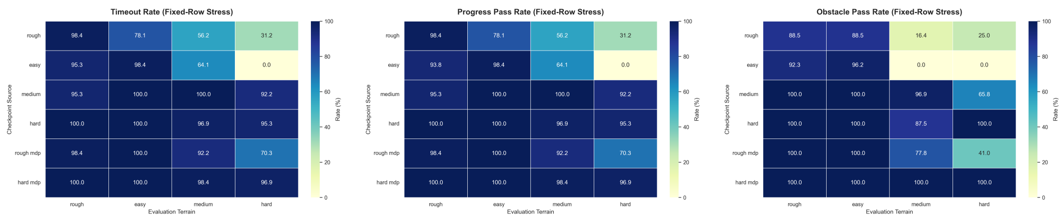
4.2 Parkour training

Figure 1 shows representative policy rollouts on rough and parkour terrains. The visual progression matches the quantitative curriculum: easy terrain mainly tests structured uneven walking, medium introduces gaps and stronger obstacles, and hard combines wider discontinuities with tighter feasible footholds.

Table 4 shows the expected terrain difficulty ordering. Easy parkour remains highly stable, medium parkour introduces a moderate degradation, and hard parkour has the lowest reward, highest fall rate,

Table 4: **Final training summary for parkour tiers.** The same 3000-iteration budget produces a clear difficulty gradient.

Metric	Easy	Medium	Hard
Mean reward	30.70	20.98	11.34
Mean episode length	983.2	988.1	941.7
Timeout rate	96.00%	93.25%	89.08%
Fall rate	4.00%	6.75%	10.92%
XY tracking error	0.3050	0.3590	0.4407
Yaw tracking error	0.6067	0.8445	1.0088
Curriculum level	5.7907	5.8001	5.6389



(a) Timeout (b) Progress (c) Obstacle

Figure 2: **Stress cross-terrain heatmaps.** Survival and progress show similar transfer patterns, while obstacle crossing exposes terrain-specific failures that timeout alone can hide.

and largest velocity tracking error. The curriculum scalar is useful within each terrain preset but should not be interpreted as the same absolute physical difficulty across different terrain generators.

4.3 Cross-terrain generalization

The 4-by-4 cross-terrain evaluation loads rough/easy/medium/hard checkpoints and evaluates each on rough/easy/medium/hard play environments. Random cross-terrain evaluation samples terrain type and difficulty inside the terrain grid, while fixed-row stress evaluation uses the highest difficulty row for each terrain preset. I report three metrics: timeout rate, traversal-progress rate, and geometry-aware obstacle-crossing rate.

Figure 2 summarizes the fixed-row stress protocol. The heatmaps make the main pattern visible: rough and easy policies do not transfer reliably to medium or hard obstacles, medium training is the first setting with strong hard-terrain survival, and hard training gives the most consistent source policy.

4.3.1 Timeout

Table 5 shows the random timeout evaluation results. The rough and easy checkpoints perform poorly on the medium and hard environments, while the medium and hard checkpoints remain strong across all evaluation environments. Among them, the hard checkpoint has the best overall generalization ability. Table 6 reports the timeout results under the fixed-row stress protocol. The overall trend is consistent with the random evaluation, but rough-to-easy and rough-to-medium performance drops noticeably under stress, which is likely caused by the increased terrain difficulty in the stress setting.

4.3.2 Traversal progress

Traversal progress is used as a forward-motion sanity check for the timeout metric. The full random and stress matrices are reported in Tables 13 and 14. The progress results broadly follow the timeout trend: rough and easy remain weak on harder terrain, while medium and hard preserve high forward-progress rates. The main exception is the stress easy-to-medium case, where progress is higher than timeout, indicating that some episodes still move forward before eventually failing.

4.3.3 Obstacle crossing

To improve the determinism of obstacle-crossing evaluation, I use only the fixed-row stress protocol for this metric. Table 7 reports the stress obstacle-crossing results. The overall trend is still consistent

Table 5: **Random cross-terrain timeout rate.**

Source	Eval rough	Eval easy	Eval medium	Eval hard
Rough	95.31%	98.44%	68.75%	29.69%
Easy	90.62%	100.00%	50.00%	0.00%
Medium	98.44%	98.44%	95.31%	87.50%
Hard	100.00%	100.00%	96.88%	95.31%

Table 6: **Stress cross-terrain timeout rate.**

Source	Eval rough	Eval easy	Eval medium	Eval hard
Rough	98.44%	78.12%	56.25%	31.25%
Easy	95.31%	98.44%	50.00%	0.00%
Medium	95.31%	100.00%	95.31%	92.19%
Hard	100.00%	100.00%	96.88%	95.31%

with the stress timeout results, but the obstacle-crossing success rates drop substantially for rough and easy on the medium and hard environments, and also for medium on the hard environment. This shows that avoiding falls does not necessarily mean that the robot has successfully crossed complex obstacles such as stairs and gaps. The hard checkpoint again shows the strongest overall performance in this evaluation.

5 Ablation experiments

5.1 MDP ablation

The MDP ablation evaluates whether lightweight reward shaping can improve obstacle competence without changing the robot, action space, observation space, terrain generator, or training budget. Two variants are trained: `rough_mdp` on official rough terrain and `hard_mdp` on hard parkour terrain.

Figure 3 and Tab. 8 show the training results after modifying the MDP. Most training metrics improve noticeably, especially episode length and fall rate. However, because the reward definition is different, the training reward should not be used as the sole evidence that the new policy is stronger. A fair comparison still requires evaluating the policies under the same timeout, progress, and obstacle metrics.

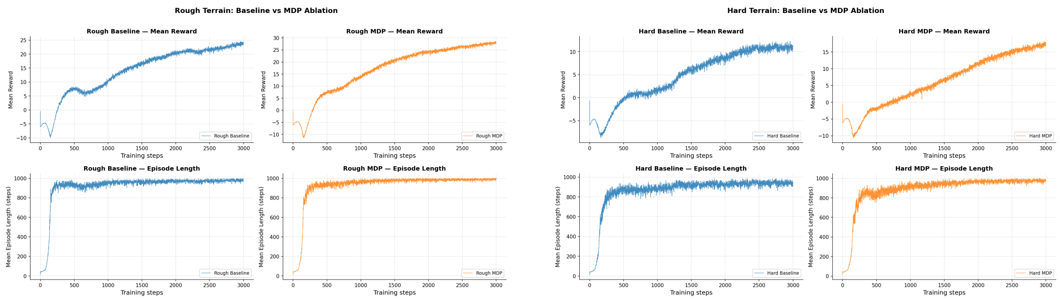
Table 9 summarizes the most diagnostic timeout results for the same cross-terrain protocols used in Section 4. The largest gain comes from Rough MDP. On hard evaluation terrain, random timeout improves from 29.69% to 70.31%, while stress timeout improves from 31.25% to 70.31%. Rough MDP also improves stress transfer to medium terrain from 56.25% to 92.19%. Hard MDP is already close to the hard baseline ceiling: it slightly improves stress hard-to-hard timeout from 95.31% to 96.88%.

The full MDP traversal-progress matrices are reported in Tables 15 and 16. They match the timeout results exactly for these four MDP-related source policies, which means the MDP-shaped policies still move forward whenever they survive. Therefore, the main MDP effect in these matrices is not a standing-still artifact; it is a genuine survival and traversal improvement, especially for rough-to-hard transfer.

Table 10 reports the fixed-row stress obstacle-crossing results for the MDP-shaped policies. The results show clear improvements for both Rough MDP and Hard MDP. In particular, Rough MDP improves obstacle-crossing success on the medium environment by 61.39 percentage points over the rough baseline and on the hard environment by 16.03 percentage points. Hard MDP improves medium obstacle crossing from 87.50% to 100.00% while keeping hard obstacle crossing at 100.00%. This suggests that the added `parkour_progress` and `foothold_safety` rewards improve obstacle competence, with the strongest effect when the baseline policy has not seen parkour terrain during training.

Table 7: **Stress cross-terrain obstacle crossing rate.**

Source	Eval rough	Eval easy	Eval medium	Eval hard
Rough	88.46%	88.46%	16.39%	25.00%
Easy	92.31%	96.15%	0.00%	0.00%
Medium	100.00%	100.00%	96.88%	65.79%
Hard	100.00%	100.00%	87.50%	100.00%



(a) Rough vs. Rough MDP (b) Hard vs. Hard MDP

Figure 3: **MDP ablation training comparison.** Reward shaping improves episode stability and final training statistics, but the policy comparison is interpreted using the shared evaluation metrics below because reward scales differ across MDPs.

5.2 ExtremeRandom OOD stress

Since both Hard and Hard MDP achieve near-perfect performance in the standard obstacle-crossing evaluation, this metric does not clearly distinguish their capabilities. Therefore, I introduce an ExtremeRandom environment that is harder than the hard obstacle environment, and compare Hard and Hard MDP under this more challenging setting. Figure 4 shows the evaluation results. Hard MDP achieves a clear advantage over Hard: the overall obstacle pass rate improves by 29.8 percentage points, gap pass rate improves by 5.5 percentage points, and stairs pass rate improves by a much larger 54.3 percentage points. These results indicate that the modified MDP also provides a substantial improvement for the hard policy.

5.3 Height-scanner ablation

The height-scanner ablation evaluates whether exteroceptive terrain information is necessary for parkour transfer. I remove the height scanner during both training and evaluation, and train two policies: `rough_no_height_scan` on official rough terrain and `hard_no_height_scan` on hard parkour terrain. The comparison uses the same fixed-row stress protocol as the main cross-terrain evaluation.

Tables 11 and 12 show a non-monotonic effect of removing the height scanner. On rough, easy, and medium stress terrain, the no-scan policies do not consistently perform worse. Rough no scan even outperforms the rough baseline on these three environments in both timeout and obstacle-crossing rate, including an increase in medium stress timeout from 56.25% to 90.62%. Hard no scan also remains close to Hard on rough and easy terrain, and preserves the same 96.88% medium timeout rate, although its medium obstacle-crossing rate drops substantially from 87.50% to 40.43%.

This result suggests that height-scan sensing is not uniformly beneficial for every moderate-terrain survival metric. On simpler or moderately difficult terrains, removing the scanner may reduce observation complexity and encourage a more conservative blind gait, which can be sufficient for staying upright. However, this interpretation does not mean that height sensing is unnecessary for parkour. On hard stress terrain, both no-scan policies collapse completely: timeout, traversal progress, and obstacle crossing all drop to 0.00%. Thus, the height scanner appears most important when the policy must adapt to hard geometric discontinuities such as difficult gaps and stairs, where proprioception alone is insufficient.

Table 8: **Final training summary for MDP ablation.** Rough and hard baselines are compared with their corresponding reward-shaped MDP variants under the same 3000-iteration budget.

Metric	Rough	Rough MDP	Hard	Hard MDP
Mean reward	24.01	28.25	11.34	17.93
Mean episode length	984.5	1000.0	941.7	991.2
Timeout rate	94.62%	97.85%	89.08%	95.29%
Fall rate	5.41%	2.15%	10.92%	4.71%
XY tracking error	0.3407	0.3718	0.4407	0.4274
Yaw tracking error	0.7501	0.6676	1.0088	0.9165
Curriculum level	5.7961	5.705	5.6389	5.769

Table 9: **MDP ablation timeout summary.** Selected columns from the random and fixed-row stress cross-terrain evaluations.

Source	Random hard	Stress medium	Stress hard	Stress hard Δ (pp)
Rough	29.69%	56.25%	31.25%	–
Rough MDP	70.31%	92.19%	70.31%	+39.06
Hard	95.31%	96.88%	95.31%	–
Hard MDP	92.19%	98.44%	96.88%	+1.57

6 Discussion

The results show that terrain exposure is the main driver of parkour robustness. Flat and rough policies validate the inherited locomotion stack, but rough and easy policies do not transfer reliably to medium or hard obstacles. Medium parkour is the first completed tier with strong hard-terrain survival transfer, while the obstacle metric still exposes limited hard-gap competence. Hard parkour remains the strongest source policy.

The MDP ablation shows a complementary effect of reward design. Rough MDP substantially improves hard-terrain transfer without parkour terrain exposure, suggesting that the inherited velocity-tracking reward alone underuses available terrain information. On hard terrain, the additional foothold-safety cost makes the policy more conservative: it reduces raw forward speed but improves tracking precision and out-of-distribution obstacle crossing.

The height-scanner ablation separates moderate-terrain survival from geometry-aware obstacle crossing. Removing height-scan observations can preserve, and in some rough-policy cases even improve, timeout and obstacle-crossing rates on rough, easy, and medium stress terrain. A plausible explanation is that the no-scan policies learn a simpler conservative blind gait that is sufficient for moderate terrain and avoids potential over-reaction to height-scan inputs. This advantage is limited: on hard stress terrain, both no-scan policies fail completely. The scanner is therefore not uniformly helpful for every survival-style metric, but it is necessary for the hardest gap and stairs cases where proprioception alone does not provide enough terrain information.

The main limitation is metric semantics. Timeout measures survival, traversal progress checks forward movement, and obstacle crossing checks base-level boundary completion. These are sufficient for the current terrain generator, but they do not directly measure foot-contact quality, exact foothold placement, or mesh-level contact safety. Future work should therefore add stricter contact-aware metrics, broader out-of-distribution obstacle layouts, and optionally motion-prior methods such as AMP/SMP-style regularization.

7 Conclusion

This project implements a reproducible parkour locomotion pipeline for Unitree G1 in Isaac Lab, including custom terrain tiers, PPO training, and timeout/progress/obstacle evaluation. The main empirical findings are:

- easy, medium, and hard terrain form a clear difficulty gradient, and hard training gives the strongest overall source checkpoint;

Table 10: **MDP ablation obstacle-crossing comparison.** Obstacle crossing is evaluated under the fixed-row stress protocol and only includes supported gap and stairs terrain groups.

Source	Eval rough	Eval easy	Eval medium	Eval hard
Rough	88.46%	88.46%	16.39%	25.00%
Rough MDP	100.00%	100.00%	77.78%	41.03%
Hard	100.00%	100.00%	87.50%	100.00%
Hard MDP	100.00%	100.00%	100.00%	100.00%

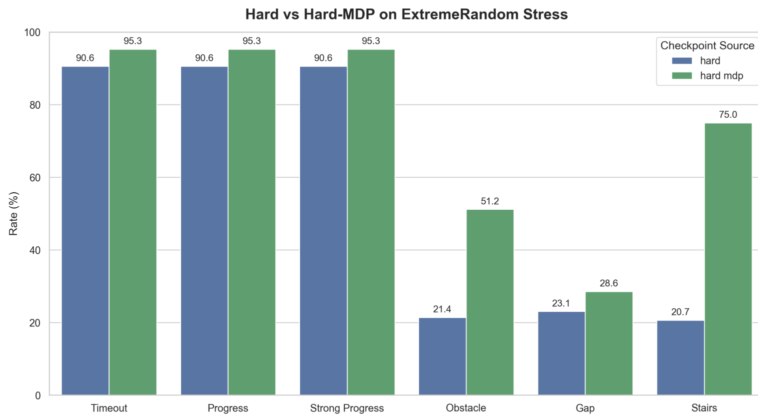


Figure 4: **ExtremeRandom hard vs. hard-MDP comparison.** Hard MDP improves survival and obstacle crossing on a harsher evaluation-only terrain distribution.

- medium training transfers well by survival metrics, but obstacle crossing shows that hard gaps still require harder terrain exposure;
- MDP reward shaping improves rough-to-hard timeout from 29.69% to 70.31% without parkour terrain exposure, while Hard MDP further improves precision and ExtremeRandom obstacle crossing;
- height-scan observations are not uniformly beneficial on moderate stress terrain, but they are critical for hard parkour: removing them drops hard-terrain stress timeout, traversal progress, and obstacle crossing to 0.00% for both rough and hard no-scan policies.

Together, these results support the feasibility of using Isaac Lab's G1 rough locomotion stack as a base for structured humanoid parkour and provide a reproducible foundation for further navigation-style or foothold-aware extensions.

References

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[3] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. URL `https://arxiv.org/abs/1707.06347`. 1

A Full traversal-progress results

Tables 13 and 14 report the full traversal-progress matrices for the four main source policies. Tables 15 and 16 report the corresponding matrices for the MDP ablation.

Table 11: **Height-scanner ablation timeout rate under fixed-row stress.** No-scan policies remove terrain height-scan observations during both training and evaluation.

Policy	Eval rough	Eval easy	Eval medium	Eval hard
Rough	98.44%	78.12%	56.25%	31.25%
Rough no scan	100.00%	100.00%	90.62%	0.00%
Hard	100.00%	100.00%	96.88%	95.31%
Hard no scan	98.44%	100.00%	96.88%	0.00%

Table 12: **Height-scanner ablation obstacle-crossing rate under fixed-row stress.** Obstacle crossing is evaluated on supported gap and stairs terrain groups.

Policy	Eval rough	Eval easy	Eval medium	Eval hard
Rough	88.46%	88.46%	16.39%	25.00%
Rough no scan	100.00%	100.00%	29.17%	0.00%
Hard	100.00%	100.00%	87.50%	100.00%
Hard no scan	100.00%	100.00%	40.43%	0.00%

Table 13: **Random cross-terrain traversal-progress rate.**

Source	Eval rough	Eval easy	Eval medium	Eval hard
Rough	95.31%	98.44%	68.75%	29.69%
Easy	89.06%	100.00%	50.00%	0.00%
Medium	98.44%	98.44%	95.31%	87.50%
Hard	100.00%	100.00%	96.88%	95.31%

Table 14: **Stress cross-terrain traversal-progress rate.**

Source	Eval rough	Eval easy	Eval medium	Eval hard
Rough	98.44%	78.12%	56.25%	31.25%
Easy	93.75%	98.44%	64.06%	0.00%
Medium	95.31%	100.00%	100.00%	92.19%
Hard	100.00%	100.00%	98.44%	96.88%

Table 15: **MDP ablation random traversal-progress rate.**

Source	Eval rough	Eval easy	Eval medium	Eval hard
Rough	95.31%	98.44%	68.75%	29.69%
Rough MDP	100.00%	100.00%	76.56%	70.31%
Hard	100.00%	100.00%	96.88%	95.31%
Hard MDP	100.00%	100.00%	96.88%	92.19%

Table 16: **MDP ablation stress traversal-progress rate.**

Source	Eval rough	Eval easy	Eval medium	Eval hard
Rough	98.44%	78.12%	56.25%	31.25%
Rough MDP	98.44%	100.00%	92.19%	70.31%
Hard	100.00%	100.00%	96.88%	95.31%
Hard MDP	100.00%	100.00%	98.44%	96.88%

B Implementation details

All completed training runs were executed on an AutoDL Linux GPU environment with an NVIDIA GeForce RTX 3090. Isaac Lab was installed externally, while this repository was installed as an editable task package. Policies were trained with PPO through RSL-RL using the inherited G1 rough runner configuration unless otherwise stated.

Table 17 summarizes the training runs used throughout this report.

Table 17: **Training runs.** All non-flat policies in this table are trained for 3000 PPO iterations.

Experiment	Task family	Iterations
Flat baseline	official flat G1	1500
Rough baseline	official rough G1	3000
Parkour easy	custom easy parkour	3000
Parkour medium	custom medium parkour	3000
Parkour hard	custom hard parkour	3000
Rough MDP	rough terrain with MDP rewards	3000
Hard MDP	hard parkour with MDP rewards	3000
Rough no scan	rough terrain without height scan	3000
Hard no scan	hard parkour without height scan	3000

The official flat baseline uses `Isaac-Velocity-Flat-G1-v0`. The rough baseline uses `Isaac-Velocity-Rough-G1-v0` and is the direct parent of all parkour tasks. The three parkour training tasks are `Isaac-Velocity-Parkour-G1-Easy-v0`, `Isaac-Velocity-Parkour-G1-Medium-v0`, and `Isaac-Velocity-Parkour-G1-Hard-v0`; each has a corresponding play task for visualization and evaluation. The MDP variants are `Isaac-Velocity-Rough-G1-MDP-v0` and `Isaac-Velocity-Parkour-G1-Hard-MDP-v0`. The height-scanner ablation uses `Isaac-Velocity-Rough-G1-NoHeightScan-v0` and `Isaac-Velocity-Parkour-G1-Hard-NoHeightScan-v0`, with matching no-height-scan play tasks for rough, easy, medium, and hard stress evaluation. Implementation files are concentrated in the environment configuration, terrain configuration, MDP override, height-scanner ablation registration, and evaluation modules of the task package.

The height-scanner ablation results used in Tables 11 and 12 are stored in `results/metrics/height_scanner_ablation_timeout_stress_eval.csv`, `results/metrics/height_scanner_ablation_progress_stress_eval.csv`, and `results/metrics/height_scanner_ablation_obstacle_stress_eval.csv`.

C Presentation video

The presentation video is available at `https://disk.pku.edu.cn/link/AA026FD72DFAD14187A300D2DE678EFD99`.